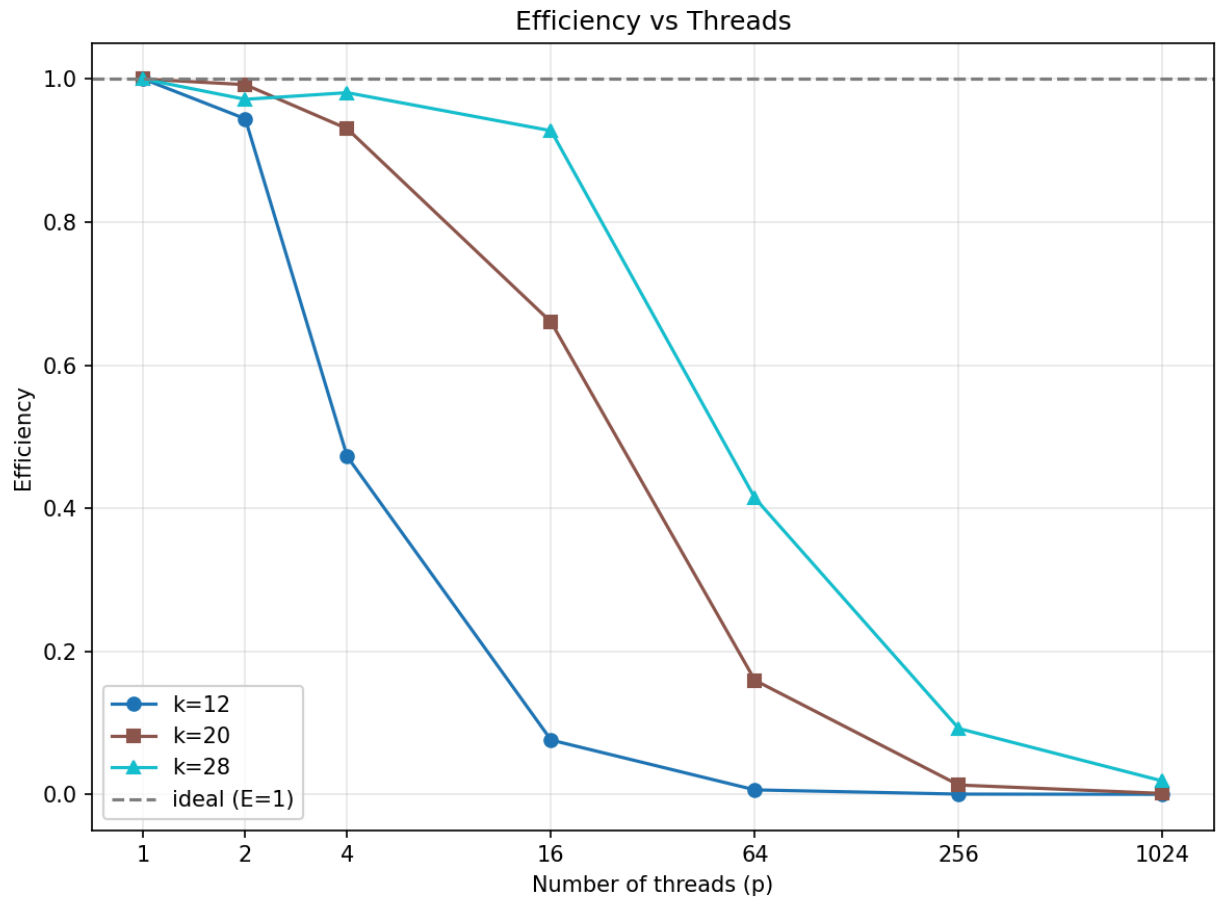
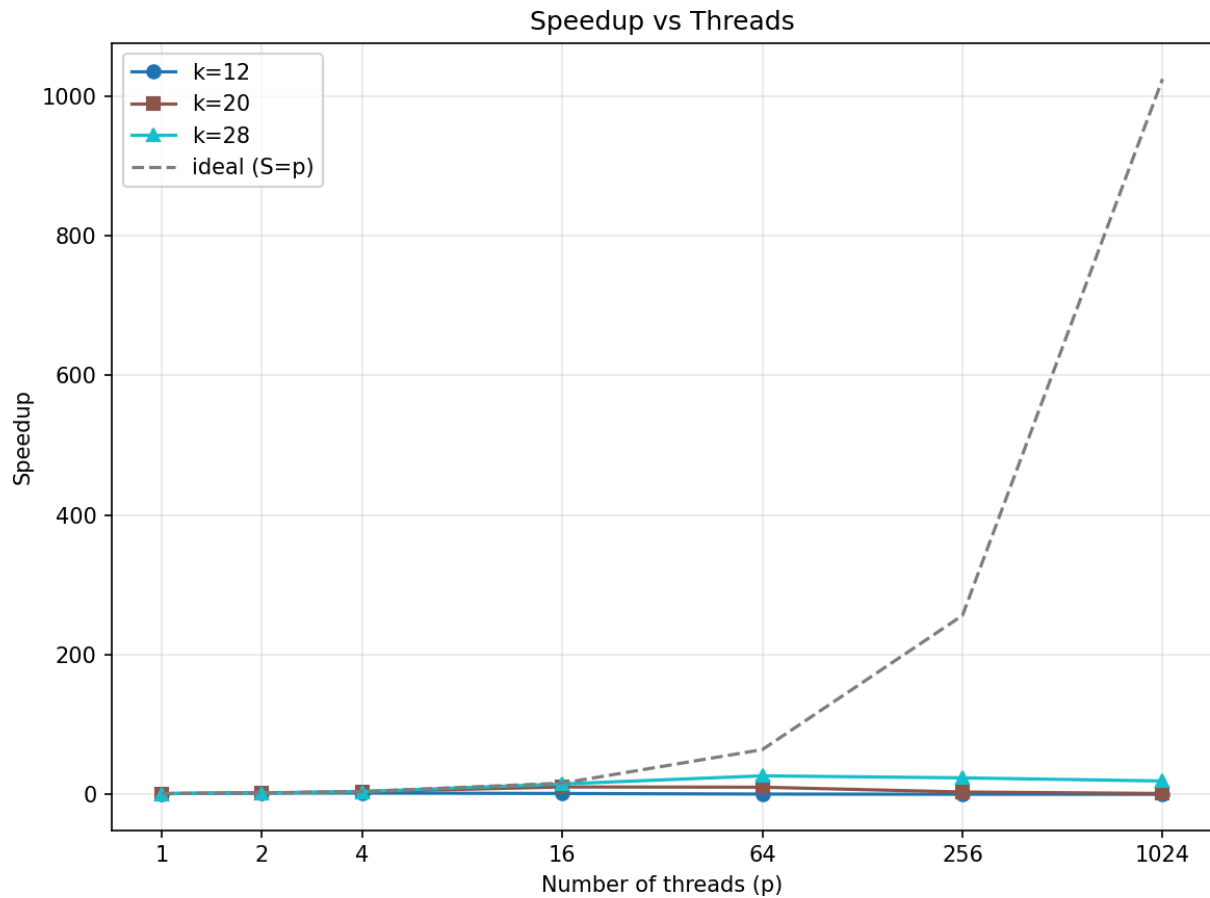


1.

```
[nnh12@grace2 HW2-735]$ ./sort_list.exe 4 1
List Size = 16, Threads = 2, error = 0, time (sec) = 0.0006,
qsort_time = 0.0000
[nnh12@grace2 HW2-735]$ ./sort_list.exe 4 2
List Size = 16, Threads = 4, error = 0, time (sec) = 0.0007,
qsort_time = 0.0000
[nnh12@grace2 HW2-735]$ ./sort_list.exe 4 3
List Size = 16, Threads = 8, error = 0, time (sec) = 0.0009,
qsort_time = 0.0000
[nnh12@grace2 HW2-735]$ ./sort_list.exe 20 4
List Size = 1048576, Threads = 16, error = 0, time (sec) = 0.0140,
qsort_time = 0.1496
[nnh12@grace2 HW2-735]$ ./sort_list.exe 24 8
List Size = 16777216, Threads = 256, error = 0, time (sec) =
0.2042, qsort_time = 2.9082
```



2.



3. My best values of k are 28 and 20. It goes from
54.31 seconds $\rightarrow$ 2.05 seconds for $k = 28$ and
0.15seconds $\rightarrow$ 0.014 seconds for $k = 20$. This is for both going from 1 to 64 threads.
This shows there is speedup when q is varied.